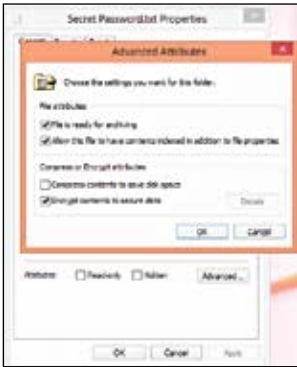


user's password. When the user changes his password, the master key is automatically decrypted and re-encrypted based on a key derived from the new password.

5. Finally, the file is protected at the 'installation' level.

This procedure provides multiple levels of security under various circumstances. One needs to know the user's password (along with all other keys) to decrypt the master key if the file is to be accessed in a different installation on the same machine – one more reason to have a strong password. When an administrator changes the password of another user, say 'UserX', the master key of UserX is not automatically decrypted/encrypted. Hence another user, even if its the administrator, can't access files encrypted by

another user (but there are workarounds for an administrator which he can set up beforehand). If the file is moved out of the machine then the user's private key, master key and the key used to encrypt the master key – are all missing. Cracking it becomes next to impossible. There's a downside though: if your installation gets corrupted or you reinstall your OS, you're locked out of your own files. To save you on such a day is this guide: <http://bit.ly/XFwygo>



Encrypting a file using EFS

Beyond EFS, NTFS has its own permission system to grant and revoke permissions for every file or folder for every user and group in the system. These options can be accessed and changed on the 'Security' tab of the file/folder properties window. This is good enough to block programs and certain files and folders from being accessed by a non-administrative user on the system. An administrator though, can change the restrictions imposed by another user.

## Registry

Yet another integral part of Windows – a central spot for all system settings, configurations and preferences – is the registry. Registry controls the behavior of the entire OS. Any setting that you change in Windows will end up changing some value in the registry. Many applications also host their settings under Registry. It controls a large part of Windows' behavior, and